

20CYS443 Biometrics and Security

Behavioural Biometrics

Signature scan

- identifies a person based on their handwriting.
- operated in two different ways:
 - 1) Static:** In this mode, users write their signature on paper, and after the writing is complete, it is digitized through an optical scanner or a camera to turn the signature image into bits.
- The biometric system then recognizes the signature analyzing its shape.

2) Dynamic: In this mode, users write their signature in a digitizing tablet, which acquires the signature in real time

- This is done by analyzing the shape, speed, stroke, pen pressure and timing information during the act of signing. Natural and intuitive, the technology is easy to explain and trust.
- Dynamic signature verification technology uses the behavioral biometrics of a hand written signature to confirm the identity of a computer user.

Keystroke scan

- Keystroke recognition is a behavioral biometric which utilizes the unique manner in which a person types to verify the identity of an individual.
- the process of measuring and assessing a typing rhythm on digital devices, including on: computer keyboards, mobile phones, and touch screen panels.
- often called “keystroke dynamics”, refers to the detailed timing information that describes exactly when each key was pressed on a digital device and when it was released as a person types.

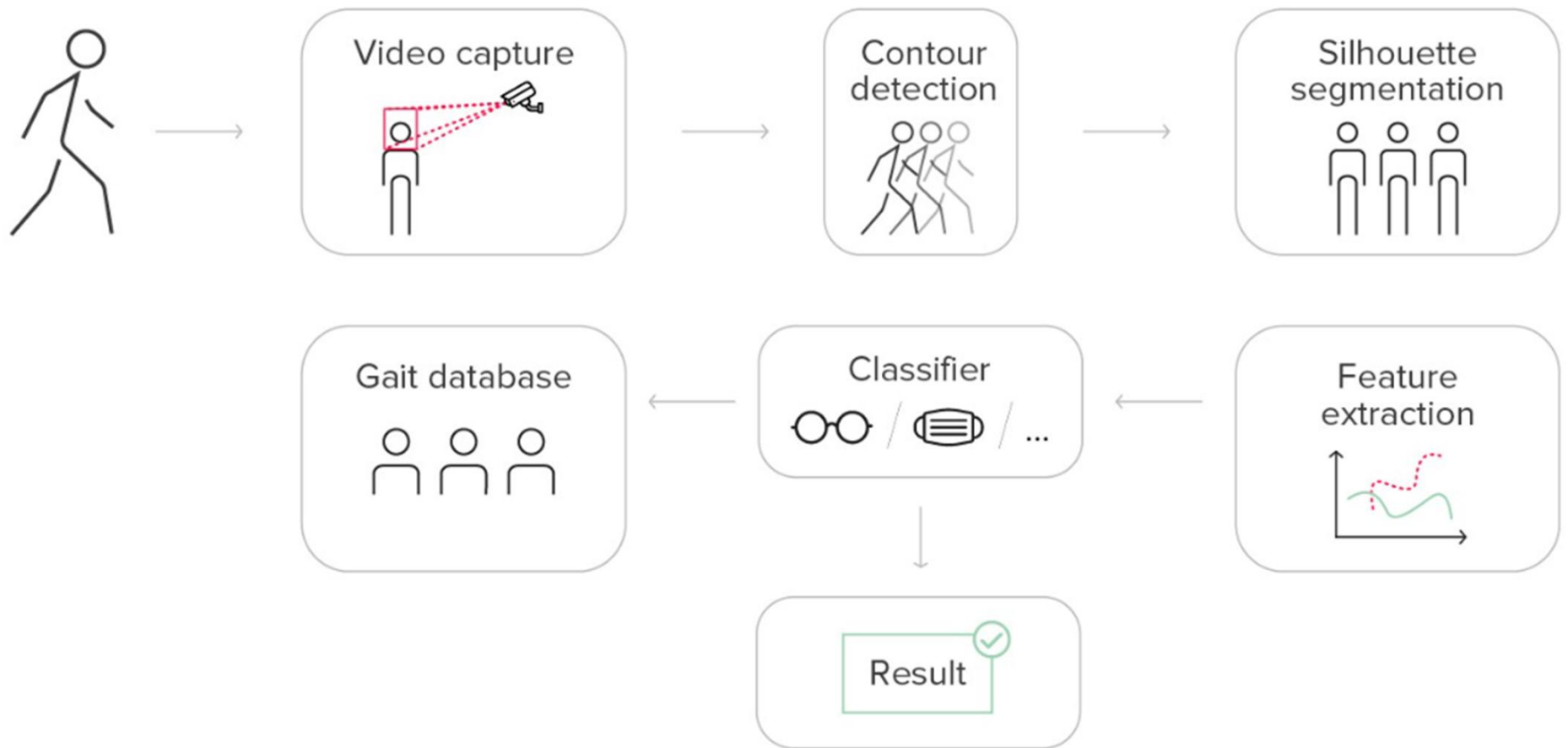
- The raw measurements used for keystroke dynamics are known as “dwell time” and “flight time”.
- Dwell time is the duration that a key is pressed, while flight time is the duration between keystrokes.
- Keystroke dynamics can therefore be described as a software-based algorithm that measures both dwell and flight time to authenticate identity.

Gait recognition

- Gait is defined as the style or manner of walking
- Gait recognition refers to the use of video of human gait, processed by computer vision methods, to recognize or to identify persons based on their body shape and walking styles.
- This is a passive mode of acquiring biometric data from a distance.

- Gait recognition technology utilizes various sources or capture devices such as cameras, motion sensors, and others to acquire data. The collected data undergo several recognition steps.
- The main algorithm recognizes the gait, processes the received data, detects contours, silhouettes, and segments individual human features. Subsequently, the feature extraction algorithm comes into effect — it distinguishes one gait from another.
- These algorithms can vary, and their requirements can also be different. For example, some algorithms are designed to process video signals, while others use data from sensors.
- A binary image of a person's silhouette is extracted from the recording and analyzed by vision-based algorithms. Silhouette segmentation simplifies the algorithm's task of processing and mapping a comprehensive image.

Gait Recognition



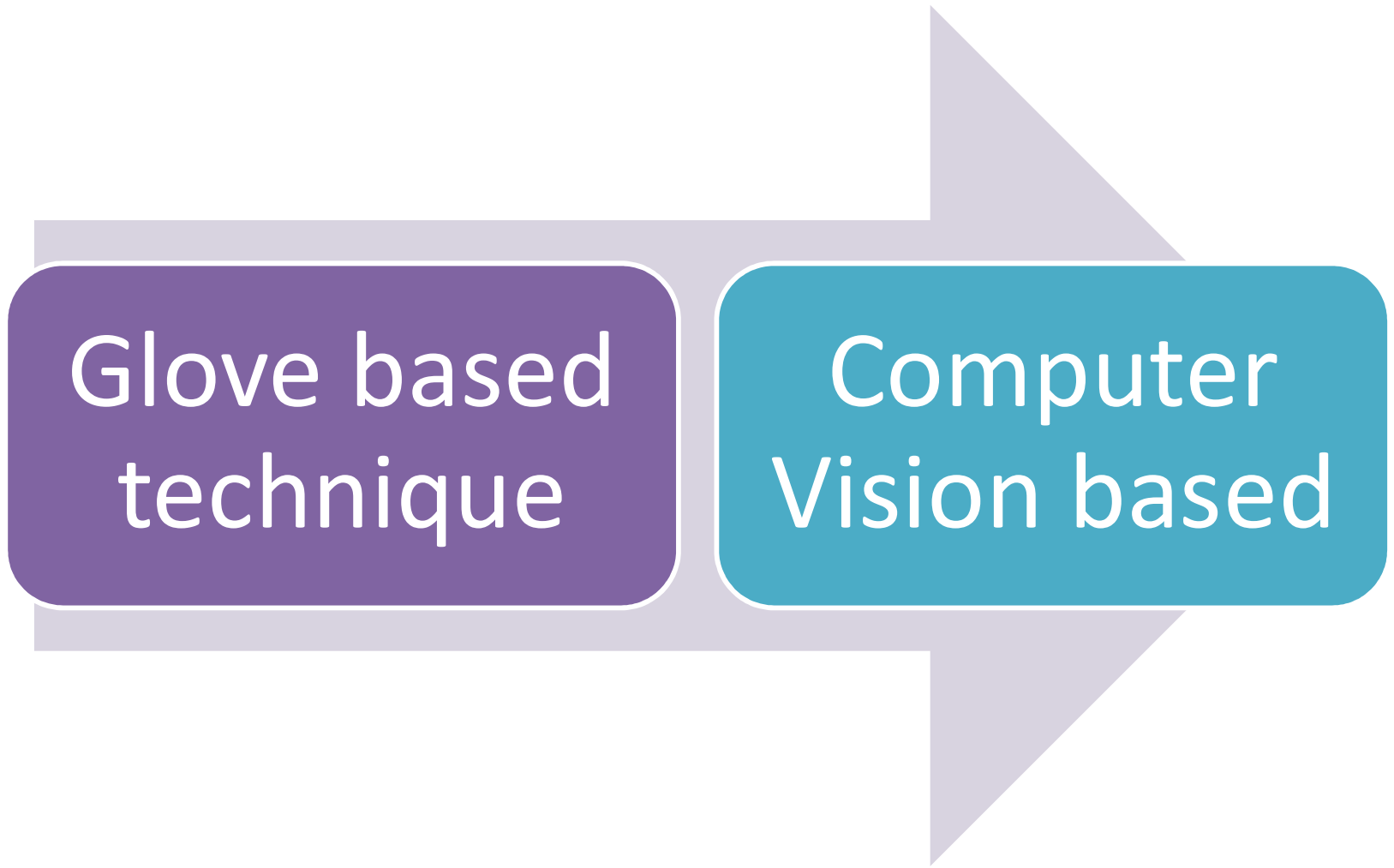
What Is Gesture Recognition?

- Gesture recognition is technology that uses sensors to read and interpret hand movements as commands.
- In the automotive industry, this capability allows drivers and passengers to interact with the vehicle — usually to control the infotainment system without touching any buttons or screens.
- Gestures originate from any bodily motion or state but commonly originate from the face or hand
- Gesture Recognition can be considered as a way for a computer to understand human body language.

Gesture Recognition - Defn

- A gesture is an action that has to be seen by someone else and has to convey some piece of information.
- The gesture is usually considered as a **movement of part of the body**, esp. a hand or the head, to express an idea or meaning.
- Gesture recognition technologies are much younger in the world of today.
- This has **minimized the need for text interfaces and GUIs** (Graphical User Interface).

Gesture Recognition- Approaches



A diagram illustrating two approaches to gesture recognition. It features a large, light purple arrow pointing to the right. Inside the arrow, there are two rounded rectangular boxes. The left box is purple and contains the text 'Glove based technique'. The right box is teal and contains the text 'Computer Vision based'.

Glove based
technique

Computer
Vision based

Gesture Recognition- Approaches

- **Glove based technique** is a well-known means of recognizing hand gestures.
- It utilizes a sensor attached to a glove that directly measures hand movements.

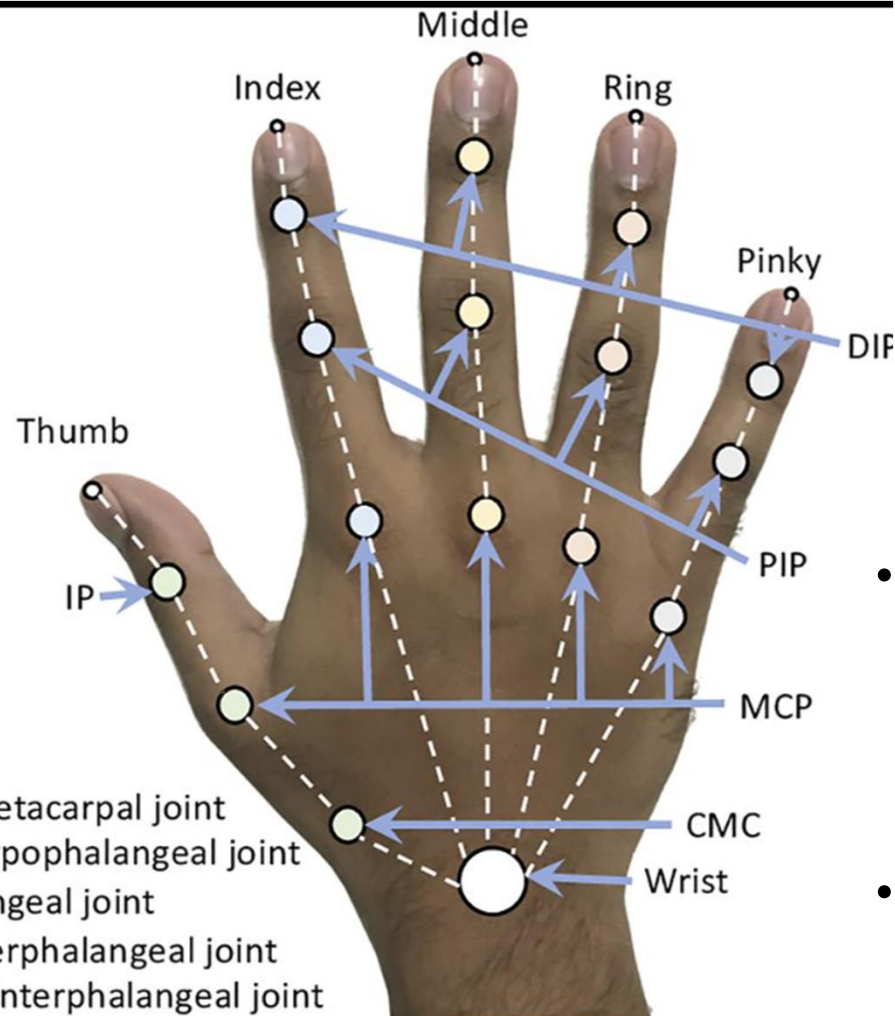
Gesture Recognition- Vision based Approach

- The sensor is a camera in vision-based gesture recognition systems
- It utilizes a sensor attached to a glove that directly measures hand movements.
- Based on feature extraction, vision-based gesture recognition systems are broadly divided into two categories,
 - appearance-based methods and
 - three dimensional (3D) hand model-based methods.

Gesture Recognition- Vision based Approach

- **Appearance-based methods**
 - Appearance-based methods utilize features of training image to model the visual appearance, and compare these parameters with the features of test image.
- **3D Hand based model:**
 - Three dimensional model-based methods rely on a 3D kinematic model, by estimating the angular and linear parameters of the model.

3D model-based hand gesture



- A **3D model-based hand gesture recognition** system represents the human hand as a **three-dimensional kinematic model** consisting of bones, joints, and fingers.
- Instead of recognizing gestures directly from images, it **estimates the hand's 3D pose** by determining the **angular (joint rotations)** and **linear (position and translation)** parameters of the hand model.

Common Sensors Used

- RGB Camera
- Stereo Camera
- Depth Camera (e.g., Microsoft Kinect, Intel RealSense)
- Leap Motion Controller
- Depth sensors improve estimation accuracy by providing explicit distance information.

- The human hand is modeled as an articulated structure with:
- **Palm:** Base of the hand.
- **Five Fingers:** Thumb, Index, Middle, Ring, and Little finger.
- **Joints:** MCP (Metacarpophalangeal), PIP (Proximal Interphalangeal), DIP (Distal Interphalangeal), and Thumb joints.
- **Degrees of Freedom (DoF):** Each joint can rotate independently, allowing the hand to perform complex gestures.

The system estimates:

- **Linear Parameters**
 - Hand position (x, y, z)
 - Translation of the hand
 - Palm location
- **Angular Parameters**
 - Joint bending angles
 - Finger orientations
 - Wrist rotation
 - Finger articulation
- Together, these parameters define the complete **3D pose** of the hand.

Input
(RGB Camera / Depth Camera / Stereo Camera)



Hand Detection & Segmentation



Feature Extraction
(Hand contour, keypoints, depth)



3D Kinematic Hand Model



Estimate Joint Angles
& Hand Position (Pose)



Gesture Classification
(Open Hand, Fist, Pointing, OK, etc.)

Feature	2D Appearance-Based	3D Model-Based
Representation	Image features	Kinematic hand model
Input	RGB image	RGB + Depth / Stereo
Parameters	Shape, contour, texture	Joint angles, rotations, translations
Viewpoint robustness	Moderate	High
Occlusion handling	Limited	Better
Computational cost	Low	High
Accuracy	Moderate	High

- **3D hand-based gesture recognition systems** are increasingly used for **authentication, access control, VR/AR, healthcare, robotics, and human-computer interaction**, they are exposed to several security and privacy vulnerabilities.
- Can you identify the security risks

Vulnerability	Description	Impact
Presentation (Spoofing) Attack	Attacker uses a 3D printed hand, silicone hand, or replayed 3D gesture to impersonate a legitimate user.	Unauthorized authentication
Replay Attack	Previously recorded RGB/depth gesture sequences are replayed to the recognition system.	Bypass authentication
Adversarial Attack	Carefully crafted perturbations cause the AI model to misclassify gestures.	Wrong gesture recognition, unauthorized actions
Sensor Spoofing	Fake depth or RGB sensor data is injected into the camera pipeline.	Incorrect hand pose estimation
Depth Map Manipulation	Modification of depth information using projected infrared patterns or fake point clouds.	Incorrect 3D model reconstruction
Occlusion Attack	Partial hiding of fingers or palm intentionally confuses the recognition algorithm.	Recognition failure or misclassification
Model Poisoning	Malicious samples are introduced during training, especially in federated learning systems.	Reduced accuracy or hidden backdoors
Backdoor Attack	A specific gesture trigger causes the model to produce attacker-chosen outputs.	Unauthorized system behavior

Data Poisoning	Incorrectly labeled gesture datasets degrade model performance.	Reduced reliability
Model Extraction Attack	Queries are used to approximate or steal the gesture recognition model.	Intellectual property theft
Membership Inference Attack	Determines whether a person's gesture data was used during training.	Privacy leakage
Gesture Injection Attack	Artificial skeletal data is injected directly into the processing pipeline.	False gesture recognition
Communication Attack	Gesture data intercepted or modified during transmission.	Data tampering and privacy breach
Denial-of-Service (DoS)	Flooding the system with gestures or malformed sensor inputs.	System unavailability

Privacy Vulnerabilities

- Even though only hand movements are captured, they can reveal sensitive information.
 - Biometric identity leakage
 - Behavioral profiling
 - Health condition inference (e.g., tremors, arthritis)
 - Finger length and palm geometry disclosure
 - User tracking across sessions
 - Activity inference from gesture patterns

AI-Specific Vulnerabilities

- Modern 3D gesture recognition uses CNNs, Transformers, or Graph Neural Networks, making them susceptible to:

1. Adversarial Examples

- Small, imperceptible perturbations to depth maps or images can fool the classifier.
- Example:
- Actual gesture:  (Thumbs Up)
- Predicted gesture:  (Stop)

2. Backdoor Attacks

- During training, the attacker inserts a trigger.
- Example:
- Whenever a small ring is worn on the finger,
- → Model always predicts "**Authorized User**".

3. Model Inversion

- An attacker reconstructs hand shape or gesture patterns from model outputs.

4. Membership Inference

- Determines whether a person's gesture data was included in the training dataset.

Sensor-Level Attacks in hand gesture recognition system

- Infrared interference with depth cameras
- Laser projection to corrupt depth measurements
- Camera blinding using strong light
- Fake skeletal joint generation
- Time synchronization attacks on multiple cameras

Network-Level Attacks

- When gesture data is sent to cloud or edge servers:
 - Man-in-the-Middle (MitM) attacks
 - Packet injection
 - Packet replay
 - Session hijacking
 - Communication eavesdropping

Hardware Vulnerabilities

- Camera tampering
- Sensor replacement
- Calibration attacks
- Firmware modification
- Physical damage or obstruction

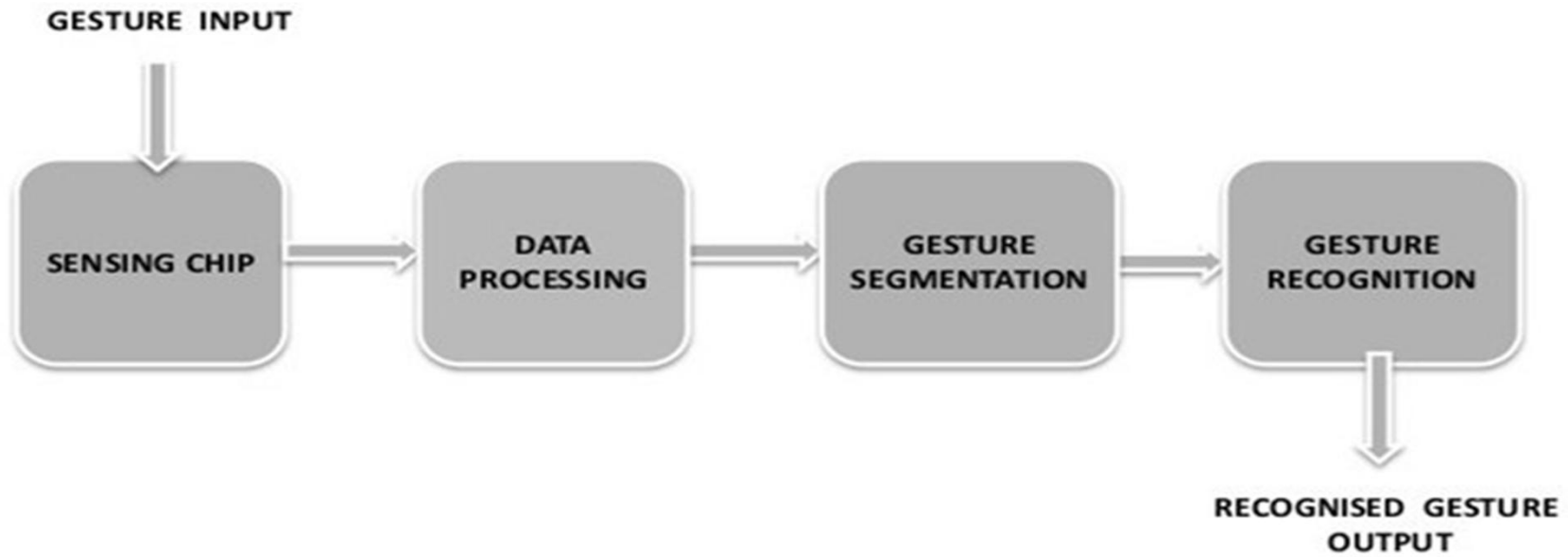
Security Countermeasures

Threat	Possible Defense
Spoofing	Liveness detection, multi-modal biometrics (hand + face), challenge–response gestures
Replay Attack	Session-specific random gesture prompts, timestamps, nonces
Adversarial Attack	Adversarial training, defensive distillation, input validation
Model Poisoning	Secure federated learning, robust aggregation (Krum, FedMedian), anomaly detection
Backdoor Attack	Backdoor detection, dataset sanitization, trigger removal
Sensor Spoofing	Sensor fusion (RGB + depth + IMU), consistency checks
Communication Attack	TLS, encryption, authentication, integrity protection
Privacy Leakage	Differential privacy, federated learning, secure enclaves, homomorphic encryption
Model Theft	Query rate limiting, watermarking, API authentication
DoS	Input validation, rate limiting, resource monitoring

Structure of Gesture Technology

- In computer interfaces, two types of gestures are distinguished:
 - 1. Offline gestures:** Those gestures that are processed after the user interaction with the object. An example is the gesture to activate a menu.
 - 2. Online gestures:** Direct manipulation gestures. They are used to scale or rotate a tangible object.

Structure of Gesture Technology



Taxonomy of gestures

- There are multiple ways to categorize hand gestures
 - Based on **observable features**
 - Based on the **interpretation**.
 - In the first category gestures are classified based on **temporal relationships**, into two types;
 - Static Gestures
 - Dynamic gestures

Taxonomy of gestures

Static hand gestures :

- ☐ Those in which the hand position does not change during the gesturing period.
- ☐ Static gestures mainly rely on the shape and flexure angles of the fingers.

Dynamic hand gestures :

- ☐ In dynamic hand gestures, the hand position changes continuously with respect to time.
- ☐ Dynamic gestures generally have three motion phases:
 - ☐ Preparation
 - ☐ stroke
 - ☐ retraction

Taxonomy of gestures

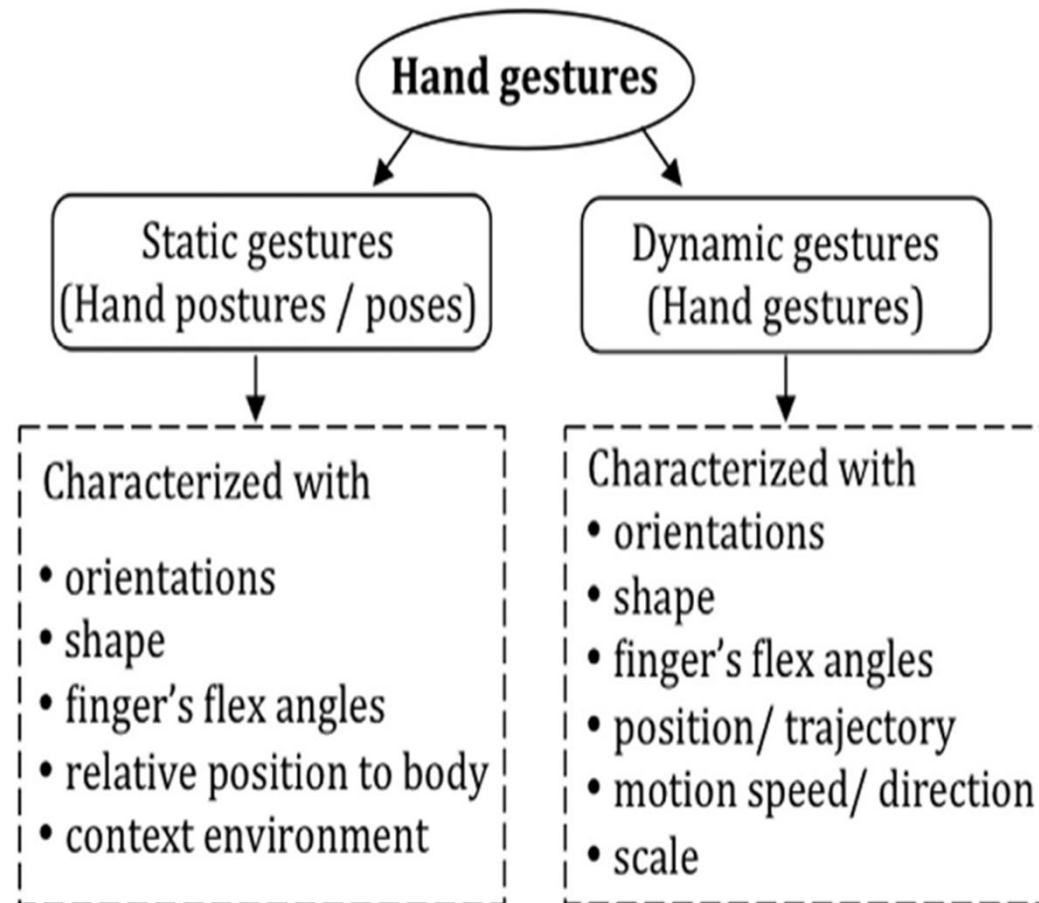


Fig. 1. Classification of hand gestures based on temporal nature. Static gestures are time independent whereas dynamic gestures are time dependent.

Taxonomy of gestures

Dynamic hand gestures :

- ❑ The message in a dynamic gesture is mainly contained in the **temporal sequence in the stroke phase**.
- ❑ Dynamic gestures rely on the **hand trajectories and orientations**, in addition to the shape and fingers' flex angles

- **Data Acquisition Or Gesture Image Collection Stage**

This is the stage for input data collection where hand, body or face gestures are recorded and classified.

- **Gesture Image Preprocessing Stage**

This step uses techniques such as edge detection, filtering, and normalization in order to capture the main gesture characteristics. It fits the input gesture into the model used for gesture recognition.

- **Image Tracking Stage**

Image tracking follows gesture image pre-processing where the sensors capture the orientation as well as the position of the object performing the gestures. This may be achieved with the help of either one or multiple trackers such as magnetic, optical, acoustic, inertial or mechanic.

The Recognition Stage

- After a successful feature extraction following the image tracking where the identified features of a gesture are stored in a system using complex neural networks, or decision trees, the command or the meaning of the gesture is declared.
- The gesture is officially recognized, and the classifier can attach every input of a test movement to its gesture class.

Levels of gesture recognition

- Gesture recognition consists of three basic levels:
- **Detection.** With the help of a camera, a device detects hand or body movements, and a machine learning algorithm segments the image to find hand edges and positions.
- **Tracking.** A device monitors movements frame by frame to capture every movement and provide accurate input for data analysis.
- **Recognition.** The system tries to find patterns based on the gathered data. When the system finds a match and interprets a gesture, it performs the action associated with this gesture. Feature extraction and classification in the scheme below implements the recognition functionality.

Applications of Gesture Recognition System

- Talking to computer



Applications of Gesture Recognition System

- **Medical Operation**

Gestures can be used to control the distribution of resources in hospitals, interact with medical instrumentation, control visualization displays, and help handicapped users as part of their rehabilitation therapy.



Applications of Gesture Recognition System

- **Gesture-based Gaming control**



In computer-vision-based, hand-gesture-controlled games, the system must respond quickly to user gestures, the “fast-response” requirement. In games, computer-vision algorithms must be robust and efficient, as opposed to applications (such as inspection systems) with no real-time requirement, and where recognition performance is the highest priority.

Applications of Gesture Recognition System

- Hand gesture to control the home appliances like MP3 player, TV etc.



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Explore about ,

Video face, Mapping the body technology.

(Assignment 2)